

James Barry

(410) 231-1200 | james.j.barry@icloud.com | College Park, MD
LinkedIn: James-J-Barry | GitHub: James-J-Barry | Portfolio Site: JimB.is

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, R, MATLAB, OCaml, Rust
Data & ML: pandas, NumPy, PyTorch, scikit-learn, SQL, Vector Embeddings, NLP, Quantum Machine Learning
Frameworks & Tools: React, Node.js, Git, CI/CD, Linux, Shell (PBS/Slurm), HPC, Apptainer/Singularity
Concepts: Data Structures & Algorithms, OOP, Parallel Computing, Containerization, Quantitative Methods

EDUCATION

University of Maryland

College Park, MD

B.S. Computer Science, Minor: Computational Finance GPA: 3.97/4.0

Graduation May 2028

- **Coursework:** Algorithms, Advanced Data Structures, Data Science, Organization of Programming Languages, Computer Systems, Linear Algebra, Probability & Statistics, Discrete Mathematics, Undergraduate Research
- **Organizations:** FIRE: Quantum Machine Learning Stream, Undergraduate Quantum Association
- **Leadership:** Logistics Director, Bitcamp Hackathon (coordinating venue, hardware, food, and utilities operations for 1000+ attendees); Education Chair, Kappa Theta Pi (leading professional development programming and technical skills development for 70+ members)

PROFESSIONAL EXPERIENCE

T. Rowe Price

Owings Mills, MD

Software Engineering Intern — AI Engineering & Application Development

May 2026 – Aug 2026

- Diagnosed and resolved a data pipeline failure in an internal AI-powered application insights platform, recovering **5,326 S3 records** and **101,726 infrastructure data points** by tracing execution across Python subprocesses, the Splunk API, and a SQLite cache layer.
- Built full-stack features integrating a React frontend with a Python backend serving **180+ API endpoints**, collaborating through GitLab CI/CD and code review in an agile team.

Parsons Corp. @ ARL DoD Supercomputing Research Center

Aberdeen Proving Ground, MD

Software Engineer & Data Scientist Intern

Jun 2025 – Aug 2025

- Built and deployed a scalable evaluation system for semantic similarity algorithms on **100,000+ Army Test Incident Reports (TIRs)**, improving evaluator accuracy and operational efficiency.
- **Improved similarity search precision by over 280%** using multi-stage LLM summarization pipelines with advanced prompt engineering and JSON parsing.
- **Cut processing time by 8×** by parallelizing TIR LLM analysis across 8-node HPC clusters using PBS batch jobs.
- Designed modular Python tooling with CLI interfaces, logging, and error handling to automate model evaluation and embedding generation; authored technical documentation and user guides for long-term maintainability.

First Year Innovation and Research Experience (FIRE)

College Park, MD

Quantum Machine Learning Researcher

Aug 2024 – Dec 2025

- Engineered a Quantum Principal Component Analysis (QPCA) algorithm utilizing Quantum Phase Estimation (QPE) to extract eigenvalues without constructing full covariance matrices.
- Validated algorithm performance achieving **86% reconstruction parity** with classical PCA on 16×16 MNIST digit images using Mean Squared Error (MSE).
- Demonstrated complexity reduction from $\mathcal{O}(N^3)$ to $\mathcal{O}(\text{poly}(\log N))$, an exponential theoretical speedup; simulated density matrix exponentiation for IonQ hardware deployment.

SELECTED TECHNICAL PROJECTS

Volatility Surface & Model-Divergence Engine

Python, NumPy/SciPy, scikit-learn, Streamlit

- Engineered a live data pipeline ingesting, cleaning, and validating **1,000+ real-time market data records**, filtering stale and illiquid inputs before feeding a numerical modeling engine.
- Built a Monte Carlo simulation engine (geometric Brownian motion, antithetic variance reduction) and a Newton-Raphson/Brent numerical solver to recover latent model parameters from observed price data, validated against closed-form benchmarks.
- Designed and cross-validated two competing statistical models of a live market surface — a constrained parametric fit and a gradient-boosted regression — reducing model-vs-market error by **roughly 90%** versus a naive baseline.
- Shipped an interactive Streamlit web application exposing the platform's output as a live, explorable 3D visualization.

High-Vis Grades for Gradescope

JavaScript, Chrome Extension API

- Engineered a Chrome Extension enhancing the Gradescope UI, providing a grade calculation feature for over **40,000 students** at UMD; implemented DOM manipulation and event listeners to inject custom HTML and dynamically display weighted course averages.